

Improving Variance Estimation for Covariate Adjustment with Binary Outcomes

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Abstract

Covariate adjustment is a general method for improving precision when estimating treatment effects in randomized trials and is recommended by the FDA in its 2023 guidance when baseline variables are prognostic for the primary outcome. We focus on a method highlighted in that guidance called “standardization” (or “g-computation”) for estimating the marginal treatment effect. We address the question of how to reliably estimate variance for binary outcomes when marginal outcome probabilities are close to 0 or 1. We propose an influence function-based leave-one-out cross-validated (IF-LOO) variance estimator for the standardized difference-in-means average treatment effect. Through simulation studies, we show that this estimator provides appropriate type-I error control and performs reliably in challenging settings where existing methods can yield inflated type-I error or fail entirely, such as when outcome events are rare or sample sizes are small. In addition to having desirable statistical properties, we derive a closed-form expression for the proposed estimator, enabling straightforward and reliable implementation by study statisticians. The robust finite-sample performance and ease of implementation suggest the IF-LOO variance estimator is a prudent default choice for standardization in clinical trials.

Keywords: Covariate Adjustment, Standardization, G-computation, Influence Function, Leave-One-Out Estimator

1 Introduction

Using baseline covariate information to increase efficiency and gain power in statistical analyses of randomized trials is a long-standing idea dating back to Fisher (1934). One method for covariate adjustment developed in the causal inference literature is called “standardization,” and should be in the toolkit of every clinical trial statistician faced with analyzing a binary endpoint. This is not only because the method was highlighted by the FDA in their covariate adjustment guidance (U.S. Food and Drug Administration, 2023), but also because it allows statisticians to perform a

covariate adjusted analysis that targets the treatment effect as a marginal risk difference. Without such methods, targeting a marginal treatment effect would only be possible in an unadjusted analysis, leaving behind any potential efficiency gains.

Standardization (Neison, 1844; J. M. Robins, 1986) is also a more robust alternative to the traditional approach of adjusting for covariates in a logistic regression and performing inference on the parameter corresponding to the treatment covariate. Standardization begins by fitting the same traditional logistic model, but only uses this regression fit as a working model. From this model, one then predicts outcomes for each subject under either active or control treatment and compares the averages of these predictions to quantify the treatment effect. By averaging across all subjects, this "standardizes" the covariate distributions when comparing the estimated means of treated and untreated subjects. Thus, this adjustment procedure can be leveraged for efficiency gains - i.e., tighter confidence intervals and more informative p-values - when the modeled covariates are prognostic. Since the probability of the outcome is marginalized in each treatment arm separately before contrasting them, this allows the analyst to chose a risk difference or ratio as opposed to the less interpretable odds ratio. Another advantage is that the treatment effect estimate is asymptotically unbiased regardless of the postulated canonical link working model (Rosenblum and van Der Laan, 2010), so the treatment effect estimated by standardization is robust to model misspecification - i.e., it is semiparametric.

While the standardization procedure is well-established and provides a straightforward way to obtain a point estimate for the treatment effect, questions remain regarding the best way to estimate the variance of the estimator (and thus construct confidence intervals). The original draft FDA guidance released in 2021 referenced a variance estimation approach that proposed using the covariance estimate from the working logistic model and applies the delta method for the risk difference transformation (Ge et al., 2011). There exist several other methods for variance estimation, including influence-function plug-in estimators (Rosenblum and van Der Laan, 2010), methods based on the limiting distribution of the standardized estimator (Ye et al., 2023), the nonparametric bootstrap (Efron, 1979), and M-estimation (Wang et al., 2023).

Since the draft guidance on covariate adjustment was published, statisticians have begun comparing the performance of some of the above estimators. Most notably, a debate was sparked in part by the theoretical work of Ye et al. (2023), which demonstrated that the variance estimation approach previously proposed by Ge et al. (2011) neglects a component of the variance, leading to underestimation. Ye et al. present a simulation study comparing the new method proposed in the paper with the method proposed in Ge et al and found that Ge et al's method had 91% coverage while Ye et al's had correct 95% coverage; however, it is unclear how applicable this result is to real-world usage, as the data-generating process for this simulation used a treatment effect corresponding to a very high odds ratio that may be unrealistic in clinical trial settings. In addition, this particular paper did not consider other variance estimators proposed in the literature. A more realistic simulation study was conducted by Liu & Xi which compared nine different variance estimators with much more comparable results among approaches, but still corroborating Ye's claims that those based the delta method have a tendency to provide anti-conservative inference (Liu and Xi, 2024). Ultimately, work such as these two papers resulted in the method in Ge et al. being removed from the final guidance (U.S. Food and Drug Administration, 2023).

However, notably absent from the literature are studies that have explicitly considered how the various variance estimators perform in scenarios with low event rates or small sample sizes, which are situations encountered in many randomized clinical trials. While exact procedures such as Fisher's (1934) or the Suissa-Shuster (1985) test are well-formulated to handle the small sample setting for binary outcomes, they cannot reap the benefits of covariate adjustment. In our own

explorations of the variance estimators for covariate adjustment when designing a phase III clinical trial, we uncovered that when event rates are low or the sample size is small, the proposed variance estimators for standardization either fail to obtain adequate coverage or suffer from computational drawbacks, which inspired us to develop a new approach. Therefore, in this manuscript, we propose a novel influence function-based leave-one-out (IF-LOO) estimate of the variance for the standardized treatment effect, which is an extension of a previously proposed influence function plug-in based variance estimator (Rosenblum and van Der Laan, 2010). This estimator draws on the history of using cross-fit and cross-validated estimators in semiparametric statistics (e.g., (van der Laan et al., 2007; Zheng and van der Laan, 2011; Newey and J. R. Robins, 2018; Chernozhukov et al., 2018; Balzer et al., 2026)). It is also inspired by the leave-one-out strategy employed in (Colantuoni and Rosenblum, 2015) which was originally used to assess whether adjusting for covariates may lead to substantial precision gains.

The outline of this paper is as follows: Section 2 reviews the methodology of standardization and describes our proposed variance approach while providing a proof of $\sqrt{n}$ consistency and asymptotic normality. Section 3 presents a simulation study in order to demonstrate adequate properties when binary event rates are low and the inappropriate coverage of other proposed influence-function based estimators of the variance. Section 4 concludes the paper with a discussion.

2 Methods

We consider the randomized control trial setting as described in (Rosenblum and van Der Laan, 2010), where the outcome of interest is binary. Suppose our trial is comprised of n participants. Let $\mathbf{X}$ denote a vector of baseline variables, A denote the treatment assignment (where $A = 1$ if the participant is assigned to the treatment arm and $A = 0$ if the participant is assigned to the control arm), and $Y \in \{0, 1\}$ denote the outcome of interest.

We adopt the potential outcomes framework (Neyman, 1923; Rubin, 1974). Each individual has potential outcomes $Y_i(1)$ and $Y_i(0)$, corresponding to the outcome under treatment and control, respectively. We assume consistency, meaning that for an individual i , their observed outcome $Y_i = Y_i(A_i)$, the potential outcome corresponding to their treatment assignment. For each individual i , we observe the tuple $O_i = (\mathbf{X}_i, A_i, Y_i)$. We assume that the tuples $O_i, i = 1, \dots, n$ are independent, identically distributed draws from an unknown data generating process $\mathbb{P}_0$. We assume that all participants are randomly assigned to the treatment or control arm with probability $\mathbb{P}_0(A = 1) = \pi_0$.

Our estimand of interest is the average treatment effect θ_{ATE} , which, under the above assumptions, is equivalent to the difference in means in the two arms:

$$\begin{aligned}\theta_{ATE} &= \mathbb{E}_{\mathbb{P}_0}[Y(1) - Y(0)] \\ &= \mathbb{E}_{\mathbb{P}_0}[Y(1)|A = 1] - \mathbb{E}_{\mathbb{P}_0}[Y(0)|A = 0] \\ &= \mathbb{E}_{\mathbb{P}_0}[Y|A = 1] - \mathbb{E}_{\mathbb{P}_0}[Y|A = 0].\end{aligned}$$

Specifically, this equality holds because the expectation can be spread across the two counterfactuals due to the linearity of expectation, A can be conditioned on because it is independent of $Y(a), \forall a \in \{0, 1\}$ due to randomization, and the last equality holds due to consistency.

2.1 Standardized Estimator

For estimation, we employ the standardized estimator with a logistic regression working model described in (Rosenblum and van Der Laan, 2010). Let $m_\beta(A, X)$ be a linear model including an

intercept, coefficient on A , and possibly other terms involving X and A (where β is the vector of coefficients). We posit the following working model:

$$\begin{aligned}\mathbb{P}(Y = 1|A, \mathbf{X}) &= \frac{1}{1 + \exp(-m_\beta(A, \mathbf{X}))} \\ &= \text{expit}(m_\beta(A, \mathbf{X})).\end{aligned}\tag{1}$$

Example 1 (Main terms model)

For concreteness, we will use a model involving A and X as main terms as a running example. More specifically, let $m_\beta(A, \mathbf{X}) = \beta_0 + \beta_A A + \beta_{\mathbf{X}}^T \mathbf{X}$. Then, our working model would be given by

$$\mathbb{P}(Y = 1|A, \mathbf{X}) = \text{expit}(\beta_0 + \beta_A A + \beta_{\mathbf{X}}^T \mathbf{X}).$$

To estimate the ATE, researchers perform the following steps:

1. Fit a logistic regression model $\hat{\mu}(a, x) = \text{expit}(m_{\hat{\beta}}(a, x))$ on the data $O_1, \dots, O_n$, estimating the coefficients $\hat{\beta}$ using maximum likelihood estimation.
2. For each individual, predict potential outcomes $\hat{Y}_i(1) = \hat{\mu}(1, X_i)$, $\hat{Y}_i(0) = \hat{\mu}(0, X_i)$.
3. The standardized estimator is given by

$$\begin{aligned}\hat{\theta}_{\text{standardized}} &= \frac{1}{n} \sum_{i=1}^n \hat{Y}_i(1) - \hat{Y}_i(0) \\ &= \frac{1}{n} \sum_{i=1}^n \hat{\mu}(1, X_i) - \hat{\mu}(0, X_i).\end{aligned}\tag{2}$$

Example 1 (Main terms model, cont.)

Let $m_{\hat{\beta}}(A, \mathbf{X}) = \hat{\beta}_0 + \hat{\beta}_A A + \hat{\beta}_{\mathbf{X}}^T \mathbf{X}$ be the linear part of the logistic regression, where the coefficients are obtained by maximum likelihood estimation in a logistic regression of the binary Y_i on A_i, X_i , and an intercept using observations $i = 1, \dots, n$. We write $\hat{\mu}(a, x) = \text{expit}(m_{\hat{\beta}}(a, x))$. Then, our estimator is given by

$$\begin{aligned}\hat{\theta}_{\text{standardized}} &= \frac{1}{n} \sum_{i=1}^n \hat{\mu}(1, X_i) - \hat{\mu}(0, X_i) \\ &= \frac{1}{n} \sum_{i=1}^n \text{expit}(m_{\hat{\beta}}(1, X_i)) - \text{expit}(m_{\hat{\beta}}(0, X_i)) \\ &= \frac{1}{n} \sum_{i=1}^n \text{expit}(\hat{\beta}_0 + \hat{\beta}_A + \hat{\beta}_{\mathbf{X}}^T \mathbf{X}_i) - \text{expit}(\hat{\beta}_0 + \hat{\beta}_{\mathbf{X}}^T \mathbf{X}_i).\end{aligned}$$

As explained in (Rosenblum and van Der Laan, 2010), the working model in Equation 1 *need not be correctly specified* in the setting of a randomized controlled trial. Even when the working model is completely misspecified, the standardized estimator is still guaranteed to be consistent for θ_{ATE} , meaning that the estimator converges to the true value as the sample size $n \rightarrow \infty$.

2.2 Variance Estimator for the Standardized Estimator

One estimator for the variance of $\hat{\theta}_{\text{standardized}}$ is presented in (Rosenblum and van Der Laan, 2010):

$$\hat{\sigma}^2 = \hat{\mathbb{V}}_{\text{standardized}}/n \quad (3)$$

where

$$\hat{\mathbb{V}}_{\text{standardized}} = \frac{1}{n} \sum_{i=1}^n \left(\left(\frac{A_i}{\pi_0} - \frac{1-A_i}{1-\pi_0} \right) (Y_i - \hat{\mu}(A_i, X_i)) + \hat{\mu}(1, X_i) - \hat{\mu}(0, X_i) - \hat{\theta}_{\text{standardized}} \right)^2 \quad (4)$$

is an estimate of the asymptotic variance of the estimator.

This variance estimator is based on estimating the variance of the *influence function* (IF) of the estimator and is consistent for the true variance of the estimator. An influence function measures the influence that a particular observation has on an estimator (Serfling, 1980; Boos and Stefanski, 2013). Any regular and asymptotically linear estimator $\hat{\theta}$ admits a unique influence function expansion (A. Tsiatis, 2006) if there exists a function φ with expectation zero such that:

$$\sqrt{n}(\hat{\theta} - \theta) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \varphi(O_i) + o_p(1). \quad (5)$$

The influence function for the standardized estimator was given in (Rosenblum and van Der Laan, 2010). As in (Rosenblum and van Der Laan, 2010), we assume there exists a maximizer β^* of the expected log-likelihood of the logistic regression, where the expectation is with respect to $\mathbb{P}_0$. Under this assumption, Rosenblum and van der Laan show that $\hat{\beta}$ converges to β^* . We write $\mu^*(a, x) = \text{expit}(m_{\beta^*}(a, x))$ to represent the logistic regression model with the true coefficients β^* . They also show that the influence function of the standardized estimator is given by

$$\varphi(O) = \left(\frac{A}{\pi_0} - \frac{1-A}{1-\pi_0} \right) (Y - \mu^*(A, X)) + \mu^*(1, X) - \mu^*(0, X) - \theta_{ATE}. \quad (6)$$

Due to the connection between the expansion in Equation 5 and the central limit theorem, the influence function φ can then be used as a means of estimating the variance of an estimator:

$$\hat{\sigma}^2(\hat{\theta}) = \frac{1}{n^2} \sum_{i=1}^n \varphi^2(O_i).$$

In the space of semiparametric models, the *efficient influence function* (EIF) is the influence function with the smallest asymptotic variance that reaches the semiparametric efficiency bound. That is,

$$\mathbb{V}(\varphi_{\text{eff}}) \leq \mathbb{V}(\varphi) \quad \forall \varphi \in \mathcal{IF}$$

where $\mathcal{IF}$ is the set of all valid influence functions within the Hilbert-space. The influence function of $\hat{\theta}_{\text{standardized}}$ in equation 3 is the efficient influence function φ_{eff} and obtains this bound if the working model is correctly specified.

Despite being grounded in rigorous asymptotic statistical theory, as we show in Section 3, this IF-based variance estimator (along with others proposed in the current literature) can underestimate variance in certain finite-sample scenarios which researchers may plausibly face when running a randomized control trial, leading to invalid type-I error control. The main situations we focus on are 1) the sample size is small or 2) the binary outcome is rare in one or both arms.

2.3 Main Result: Leave One Out EIF Variance Estimation

To overcome this finite-sample error, we propose a new variance estimator that builds upon the influence function based estimator above. This variance estimator employs a leave one out cross-validation (IF-LOO) approach to estimate the variance of the influence function.

Intuitively, one can think of the underestimation resulting from using the estimator $\hat{\sigma}^2$ as an overfitting problem. In particular, the term $\left(\frac{A_i}{p_0} - \frac{1-A_i}{1-p_0}\right)(Y_i - \hat{\mu}(A_i, X_i))$ appears in Equation 3. When the sample size is small, or there is little variation in the observed outcomes, it is easy to see that our model may overfit to our particular data, meaning our estimates of the residuals $(Y_i - \hat{\mu}(A_i, X_i))$ are too small relative to the true residual $Y_i - \mathbb{E}_{\mathbb{P}_0}[Y|A_i, \mathbf{X}_i]$. By employing leave-one-out, we prevent such overfitting by ensuring the prediction for individual i is generated using a model that was trained on all data except for that from individual i .

Let $\hat{\mu}_{-i}(a, x)$ be the predictions from a fitted working logistic regression model using all of the available data *excluding* individual i ($n - 1$ data points). Then, we define

$$\hat{\sigma}_{\text{IF-LOO}}^2 = \frac{1}{n^2} \sum_{i=1}^n \hat{\varphi}_{-i}^2, \quad (7)$$

where

$$\hat{\varphi}_{-i} = \left(\frac{A_i}{p_0} - \frac{1-A_i}{1-p_0}\right) (Y_i - \hat{\mu}_{-i}(A_i, X_i)) + \hat{\mu}_{-i}(1, X_i) - \hat{\mu}_{-i}(0, X_i) - \hat{\theta}_{\text{standardized}}.$$

In addition, we present the following theorem, stating that the proposed estimator is $\sqrt{n}$ consistent and asymptotically normal:

Theorem 1. *Under Conditions 1-4 in Appendix A.1,*

$$\sqrt{n} (\hat{\mathbb{V}}_{\text{IF-LOO}} - \mathbb{V}_0) \xrightarrow{\mathcal{D}} \mathcal{N}(0, W),$$

where $W = \text{Var}(\varphi(O)^2)$.

We present the proof of this theorem in Appendix A. We evaluate the finite sample properties in Section 3 via simulation studies and show that the estimator results in much better coverage than other influence-function based methods for variance estimation.

Example 1 (Main terms model, cont.)

For an individual i , let $m_{\hat{\beta}_{-i}}(A, \mathbf{X}) = \hat{\beta}_{0,-i} + \hat{\beta}_{A,-i}A + \hat{\beta}_{\mathbf{X},-i}^T \mathbf{X}$ be the estimated linear part of the logistic regression model using all observations excluding $(\mathbf{X}_i, A_i, Y_i)$, where the coefficients are obtained by maximum likelihood estimation in a logistic regression of Y on A, X , and an intercept. For each individual, calculate

$$\begin{aligned} \hat{\varphi}_{-i} = & \left(\frac{A_i}{p_0} - \frac{1-A_i}{1-p_0}\right) (Y_i - \text{expit}(m_{\hat{\beta}_{-i}}(A_i, \mathbf{X}_i))) \\ & + \text{expit}(m_{\hat{\beta}_{-i}}(1, \mathbf{X}_i)) - \text{expit}(m_{\hat{\beta}_{-i}}(0, \mathbf{X}_i)) - \hat{\theta}_{\text{standardized}} \end{aligned}$$

Then, our estimator is given by

$$\hat{\sigma}_{\text{IF-LOO}}^2 = \frac{1}{n^2} \sum_{i=1}^n \hat{\varphi}_{-i}^2.$$

3 Simulation Studies

3.1 Simulation Setup

To evaluate the finite-sample performance of our estimator, we perform simulation studies mimicking realistic scenarios one may expect in a randomized clinical trial. Using simulated data allows us to evaluate the performance of the different estimators against a known ground truth. We consider two hypothetical randomized control trials with two data generating processes (DGPs) to highlight the types of scenarios in which researchers would gain the most from the IF-LOO variance estimator. The first is a scenario with a moderate sample size ($N = 250$) and a small marginal placebo rate of the outcome in the population (2.5%). The second is a scenario with a smaller sample size ($N = 50$) and a larger marginal placebo rate of the outcome in the population (25%).

In each of the two scenarios, each individual observation consists of six independent covariates: four continuous and two binary. The treatment is assigned via a random coin flip. The outcome is a binary variable where the log odds is linear in all six covariates and the treatment. We consider different average treatment effects of 0%, 2.5%, 5%, 10%, and 15%. Since the ground truth for the marginal risk difference is not a parameter from the data-generating process, we utilize Gaussian quadrature to compute the true causal effect (Ocampo et al., 2026; Gauss, 1814). Each DGP is sampled 10,000 times, and each method is applied to each replicate. Explicit details on the simulation can be found in Appendix B and R code is included in the supplementary materials to foster reproducibility.

We consider two of the ATE estimators in each simulation: an unadjusted analysis and the standardized estimator. The unadjusted analysis is implemented by the difference-in-means estimator, with the variance calculated via a plug-in for the variance of a Bernoulli random variable. The standardized estimator is implemented by fitting a logistic regression working model on the data. We use a correctly specified model when implementing the standardized estimator. The reason for this is two fold. As described above in Subsection 2.2, if the working model is correctly specified, then the resulting standardized estimator has the lowest possible variance (in a semi-parametric sense) – thus, using a correctly specified model is optimal in the sense of achieving the most precision gains possible out of covariate information. At the same time, scenarios in which researchers use a correctly specified working model is precisely when we expect the most overfitting. Thus, using a correctly specified working model is when we would expect traditional estimators of the variance to perform the worst.

As benchmark methods, we consider three estimators of the variance for the standardized estimator popular in the literature: the influence-function based estimator described in (Rosenblum and van Der Laan, 2010) (which we refer to as the “IF plug-in” estimator), the analysis of heterogeneous covariance (ANHECOVA) estimator as implemented in the RobinCar2 package (Li et al., 2026), and the nonparametric bootstrap (over 1000 bootstrap samples) (Efron, 1979). To evaluate the performance of the estimators, we consider the bias of the variance estimator as well as the coverage of the normal-based 95% confidence interval.

3.2 Simulation Results

The main results of the simulation studies are presented in Table 1 and Table 2, which display the empirical coverage of the nominal 95% confidence interval across the five DGPs for each scenario.

We first consider moderate sample size scenario of $N = 250$ with a low placebo rate of 2.5% (Figure 1). IF-LOO achieves coverage near the nominal level across all DGPs, as does the unad-

Table 1: Simulation results ($N = 250$, placebo rate = 2.5%)

ATE	Metric	Method				
		IF-LOO	IF Plug-In	RobinCar2	Bootstrap	Unadjusted
0%	SE	0.0207	0.0207	0.0207	0.0207	0.0201
	Est. SD	0.0198	0.0171	0.0172	0.0230	0.0194
	Type I Error	5.52	11.83	11.80	3.56	5.29
2.5%	SE	0.0237	0.0237	0.0237	0.0237	0.0243
	Est. SD	0.0229	0.0207	0.0208	0.0247	0.0236
	Coverage	94.35	91.14	91.43	95.64	94.14
5%	SE	0.0258	0.0258	0.0258	0.0258	0.0275
	Est. SD	0.0256	0.0237	0.0238	0.0265	0.0271
	Coverage	94.57	92.15	92.37	95.53	94.17
10%	SE	0.0298	0.0298	0.0298	0.0298	0.0323
	Est. SD	0.0300	0.0283	0.0284	0.0301	0.0324
	Coverage	94.65	92.99	93.27	95.10	95.00
15%	SE	0.0333	0.0333	0.0333	0.0333	0.0368
	Est. SD	0.0333	0.0317	0.0319	0.0334	0.0365
	Coverage	94.83	93.56	93.67	94.98	94.24

Note: SE is the empirical standard deviation of the ATE point estimates over replicates. Type I error is the proportion of replicates ($\times 100$) for which the null hypothesis of no treatment effect is rejected at $\alpha = 0.05$. Coverage is the proportion of replicates ($\times 100$) for which the nominal 95% CI contained the true ATE. A small number of replicates (0.05% for ATE = 2.5% and 0.51% for ATE = 0%) were excluded for all estimators due to errors thrown by the RobinCar2 package for variance estimation; replicates were excluded jointly across estimators to avoid conferring an advantage to any estimator by discarding potentially difficult replicates.

justed analysis. Bootstrap exhibits slight overcoverage at low true ATE values, while IF Plug-In and RobinCar2 continue to undercover at low true ATE values, though the degree of undercoverage is substantially attenuated relative to the $N = 50$ scenario and both converge toward the nominal level as the true ATE increases. This attenuation is expected: as the sample size grows, the overfitting bias in the plug-in variance estimator shrinks, and the advantage of the LOO correction diminishes. Nevertheless, the persistence of undercoverage for IF Plug-In and RobinCar2 even at $N = 250$ suggests that the correction offered by IF-LOO remains practically relevant at moderate sample sizes.

We next consider the more challenging scenario: a small sample size of $N = 50$ with a placebo rate of 25% (Figure 2). In this setting, IF Plug-In and RobinCar2 exhibit substantial undercoverage across all DGPs, with type-I error coverage as high as 18% at the null. Coverage of the confidence intervals resulting from these two estimators improve only modestly as the true ATE increases, never approaching the nominal level within the range of DGPs considered. This is consistent with the theoretical motivation for the IF-LOO estimator: in small samples with a correctly specified working model, the plug-in variance estimator is downward biased due to overfitting of the outcome model, and this bias is severe enough to meaningfully distort inference. In contrast, the bootstrap estimator achieves coverage near the nominal level across all DGPs. The IF-LOO and unadjusted

Table 2: Simulation results ($N = 50$, placebo rate = 25%)

ATE	Metric	Method				
		IF-LOO	IF Plug-In	RobinCar2	Bootstrap	Unadjusted
0%	SE	0.116	0.116	0.116	0.116	0.124
	Est. SD	0.111	0.084	0.087	0.129	0.120
	Type I Error	6.89	18.35	17.18	4.88	6.45
2.5%	SE	0.116	0.116	0.116	0.116	0.124
	Est. SD	0.112	0.086	0.088	0.130	0.123
	Coverage	93.51	83.17	84.16	95.36	93.82
5%	SE	0.118	0.118	0.118	0.118	0.128
	Est. SD	0.112	0.087	0.090	0.130	0.124
	Coverage	93.37	83.69	84.78	95.13	93.83
10%	SE	0.117	0.117	0.117	0.117	0.131
	Est. SD	0.114	0.090	0.092	0.130	0.057
	Coverage	93.88	85.85	86.83	95.42	93.67
15%	SE	0.117	0.117	0.117	0.117	0.129
	Est. SD	0.151	0.092	0.095	0.128	0.060
	Coverage	93.69	86.88	87.78	95.15	93.77

Note: SE is the empirical standard deviation of the ATE point estimates over replicates. Type I error is the proportion of replicates ($\times 100$) for which the null hypothesis of no treatment effect is rejected at $\alpha = 0.05$. Coverage is the proportion of replicates ($\times 100$) for which the nominal 95% CI contained the true ATE. A small number of replicates (between 0.05% and 0.6%) were excluded for all estimators from all DGPs due to errors thrown by the RobinCar2 package; replicates were excluded jointly across estimators to avoid conferring an advantage to any estimator by discarding potentially difficult replicates.

analysis both exhibit modest undercoverage, with coverage remaining stable between 93% and 94% across the range of true ATE values.

Taken together, the simulation results suggest that IF-LOO offers a meaningful improvement over the IF plug-in and RobinCar2 variance estimators in finite samples, particularly in the small sample, higher placebo rate regime where overfitting is most pronounced. Bootstrap performs comparably to IF-LOO in terms of coverage. However, as we discuss further in [Section 4](#), the nonparametric bootstrap is not deterministic (i.e., it has Monte Carlo error). Running the nonparametric bootstrap with 1,000 bootstrap samples in the same data set multiple times can result in different estimates for the standard error. Increasing the number of bootstrap samples can decrease this source of error - however, it comes at increased computational time. Furthermore, working logistic regression models can fail to fit for some bootstrap samples where complete separation (all 0s or 1s in a particular covariate strata) occurs. This leads to having to throw out some bootstrap samples, leading to an invalid bootstrap distribution. Because the IF-LOO variance has a closed-form expression, it avoids the computational instability and sampling errors associated with the bootstrap.

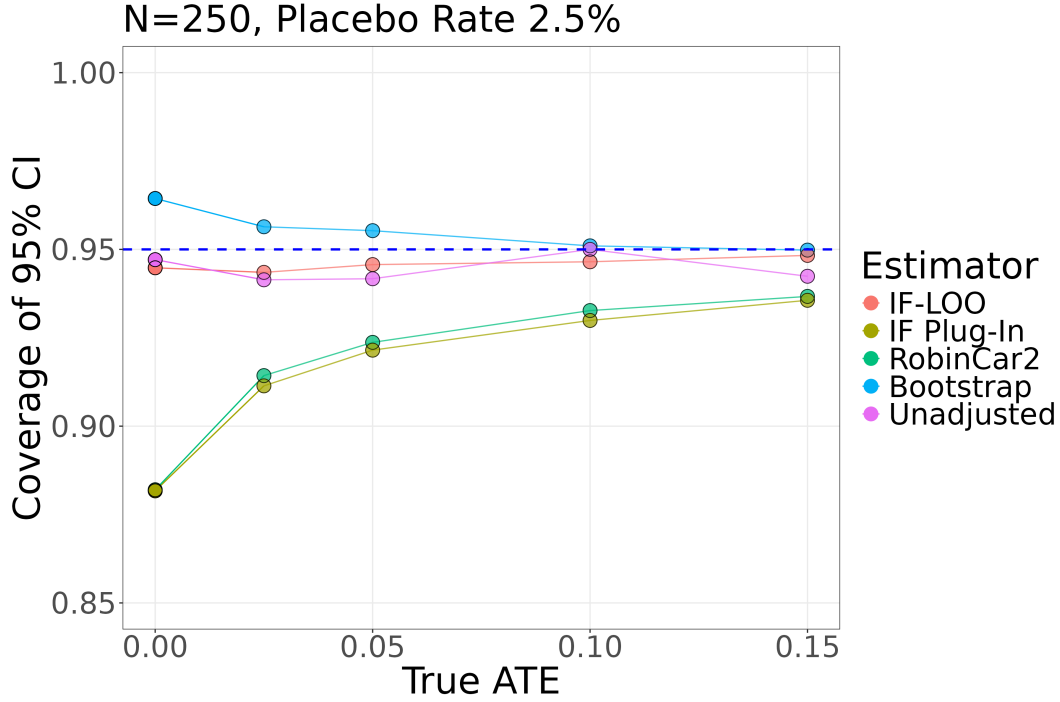


Figure 1: Simulation results for $N = 250$ with a placebo rate of 2.5%. The x -axis displays the true ATE and the y -axis displays the empirical coverage of the nominal 95% confidence interval across 10,000 replicates, with the dashed blue line indicating the nominal 0.95 level. A small number of replicates (0.05% for ATE = 2.5% and 0.51% for ATE = 0%) were excluded for all estimators due to numerical instability in matrix inversion during RobinCar2 variance estimation; replicates were excluded jointly across estimators to avoid conferring an advantage on RobinCar2 by discarding potentially difficult replicates. IF-LOO and the unadjusted analysis achieve coverage near the nominal level across all DGPs, while IF Plug-In and RobinCar2 exhibit undercoverage at low true ATE values, converging toward nominal coverage as the true ATE increases (but still undercovering). Bootstrap exhibits slight overcoverage at low true ATE values but returns to nominal levels at moderate and larger ATEs.

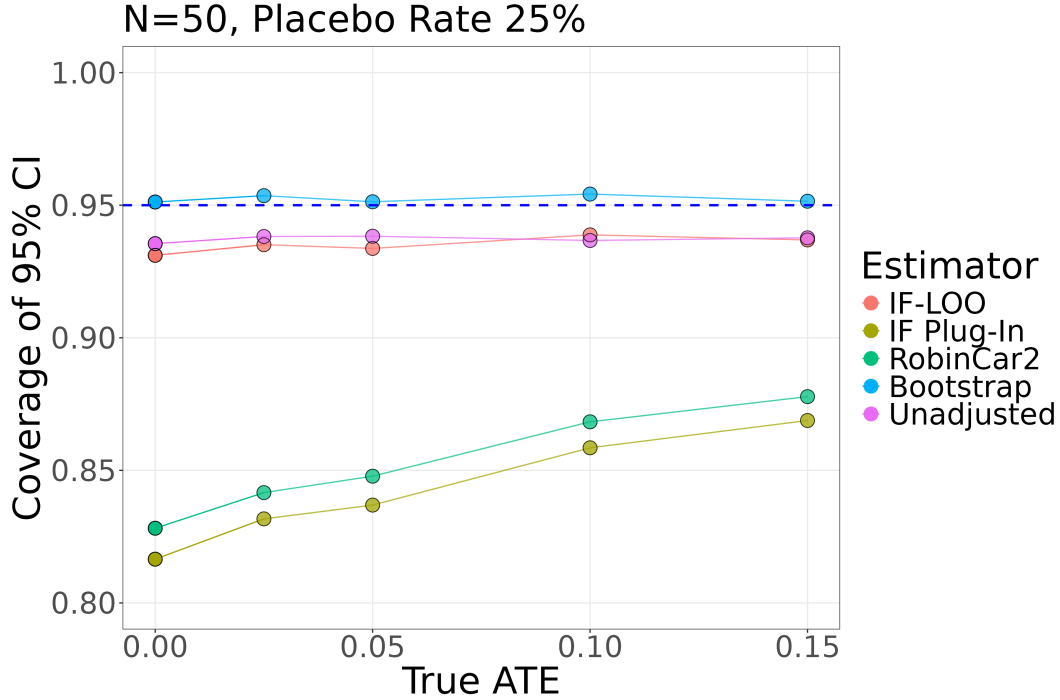


Figure 2: Simulation results for $N = 50$ with a placebo rate of 25%. The x -axis displays the true ATE and the y -axis displays the empirical coverage of the nominal 95% confidence interval across 10,000 replicates, with the dashed blue line indicating the nominal 0.95 level. A small number of replicates (between 0.05% and 0.6%) were excluded for all estimators due to numerical instability in matrix inversion during RobinCar2 variance estimation; replicates were excluded jointly across estimators to avoid conferring an advantage on RobinCar2 by discarding potentially difficult replicates. Bootstrap achieves coverage near the nominal level across all DGPs. IF-LOO and the unadjusted analysis exhibit modest undercoverage, remaining stable across true ATE values. IF Plug-In and RobinCar2 exhibit substantial undercoverage across all DGPs, with coverage increasing toward but not reaching the nominal level as the true ATE increases.

4 Discussion

Traditionally, covariate adjustment for binary outcomes involved reporting the coefficient of treatment from a logistic regression model that incorporates treatment and covariates as an estimate for a treatment effect. However, this approach complicates the interpretation of the treatment effect due to the non-collapsibility of the odds ratio (Prentice and Gloeckler, 1978; Greenland et al., 1999). Non-collapsibility leads to the coefficients in the unadjusted and adjusted logistic regression models to actually target different estimands. This means that the statistician’s choice of model specification will change the targeted treatment effect, which is not a desirable feature. Therefore, the simplest reason to prefer the marginal treatment effect to other conditional treatment effects is that, for a given dataset, there is only one marginal treatment effect, while there are many conditional effects, since there is an infinite choice of potential adjustment models. Marginal effects are also aligned with the treatment effect that is targeted when comparing outcomes between the arms of a study. The move towards targeting marginal effects on the risk difference scale in clinical trials is therefore welcomed and standardization provides a covariate-adjusted avenue for efficiently targeting these effects.

In this paper, we demonstrate the utility of a LOO influence function based variance estimator for the risk difference of a binary endpoint estimated via standardization. We demonstrate that the approach is well grounded in the asymptotic theory of influence functions and prove consistency of the proposed IF-LOO variance estimator. Via simulations, we demonstrate that in small samples and those with low event rates the proposed variance estimator provides more reliable coverage and proper type-I error control as compared to standard methods proposed in the literature. Standard methods have a tendency to overfit the data in these situations (especially when there are many covariates relative to the number of observations). In small samples, even a few covariates can lead to overfitting and improper model specification. Based on the semi-parametric theory of the efficient influence function, the correctly specified model is the one that achieves the minimum proper variance (i.e., the semiparametric efficiency bound). The IF-LOO estimator seems to perform best especially in these scenarios. In large samples with non-rare event rates, the differences between the various variance estimators disappears and the issue is less of a concern.

The IF-LOO approach has a number of advantages. First, in clinical trial programming implementations where double-coding and precise matching of results is common practice, the LOO approach enables unambiguous exact matching because it does not involve any repeated sampling. Also, machine learning algorithms (e.g., ridge, lasso, etc.) could be easily employed for the plug-in predictions since the variance is influence function based, although as discussed above, more sophisticated functional forms run the risk of more overfitting, which is already a problem in these scenarios. Cross-validation or sample splitting is often necessary when employing machine learning algorithms, which while theoretically well-principled, has been shown to lead to convergence issues in finite-samples (Shao et al., 2026). Operationally, clinical trial analyses must often be double-coded, and a LOO based estimator has the luxury of being deterministic given the data - i.e., there is no random sampling involved, as with cross-validation). In general, this work helps us observe that with dichotomized data, unforeseen problems in covariate adjustment are more likely to arise. We also observed that with binary covariates, the under-coverage of overfit models tends to be more of an issue. Hence, we considered a mixture of binary and uniform distributed variables in our simulation study.

A number of avenues for future research are possible. Firstly, finite sample corrections (A. A. Tsiatis et al., 2008) on the asymptotic-based variance estimates could help fix the under coverage of these methods revealed in our simulation studies. Our approach is grounded in super-population inference, as this is the standard inferential framework for clinical trials; however, the relationship of these methods to randomization-based inference (Abadie et al., 2020) should be explored more deeply. Magirr et al. (2025) have noted that the inferential framework and desired estimand can inform the choice of variance estimator in the context of standardization. Relatedly, exact tests may also have the potential to integrate covariate adjustment via linearization of the test statistic, as done in (Sun et al., 2025). We welcome motivated researchers to pursue these paths.

Overall, the IF-LOO variance estimator solves a number of issues currently faced in covariate adjustment with binary outcomes - especially when faced with low event rates or small sample sizes. Through both simulation and theory we demonstrate its advantageous properties and feel it a prudent default choice as a variance estimate to ensure valid inferences.

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A Proof of Theorem 1

In this appendix, we will prove [Theorem 1](#). We first lay out a few necessary assumptions:

A.1 Conditions

1. There exists a maximizer β^* of the expected log-likelihood of the logistic regression, where the expectation is with respect to $\mathbb{P}_0$.
2. Bounded propensity scores: $\eta < \pi_0 < 1 - \eta$ for $0 < \eta < 1$.
3. Stability: $\|\hat{\mu}_{-i} - \hat{\mu}\|_{L_2} = o_p(n^{-1/2})$
4. Bounded fourth moments: $\mathbb{E}_{\mathbb{P}_0}[\varphi(O_i)^4] < \infty$

The first condition is discussed in ([Rosenblum and van Der Laan, 2010](#)). The second condition is automatically met in randomized control trials where the propensity score is bounded away from 0 and 1. The third condition is automatically met when using a working logistic regression model to fit $\hat{\mu}$ when choosing a set number of covariates to adjust for that does not grow with sample size n . The fourth condition is satisfied for most reasonable probability distributions.

As in [Subsection 2.2](#), let $\mu^*(a, x) = \text{expit}(m_{\beta^*}(a, x))$ be logistic regression model with the true coefficients β^* (as shown in ([Rosenblum and van Der Laan, 2010](#)), this is guaranteed to exist under conditions 1 and 2). As discussed in section [Subsection 2.2](#), the IF of the standardized estimator is given by

$$\varphi(O) = \left(\frac{A}{\pi_0} - \frac{1-A}{1-\pi_0} \right) (Y - \mu^*(A, X)) + \mu^*(1, X) - \mu^*(0, X) - \theta_{ATE}. \quad (8)$$

Then, we can write the true asymptotic variance of the standardized estimator for the ATE $\mathbb{V}_0 = \mathbb{V}(\hat{\theta}_{\text{standardized}})$ as

$$\mathbb{V}_0 = \mathbb{E}_{\mathbb{P}_0}[\varphi(O)^2].$$

Rosenblum and van der Laan prove that the standardized estimator is asymptotically normal with variance $\mathbb{V}_0/n$ ([Rosenblum and van Der Laan, 2010](#)):

$$\sqrt{n} \left(\hat{\theta}_{\text{standardized}} - \theta_{ATE} \right) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \mathbb{V}_0).$$

In [Equation 7](#), we defined

$$\hat{\sigma}_{\text{IF-LOO}}^2 = \frac{1}{n^2} \sum_{i=1}^n \hat{\varphi}_{-i}^2.$$

For the proof below, we will be working with the IF-LOO estimator of the asymptotic variance

$$\begin{aligned} \hat{\mathbb{V}}_{\text{IF-LOO}} &= n \hat{\sigma}_{\text{IF-LOO}}^2 \\ &= \frac{1}{n} \sum_{i=1}^n \hat{\varphi}_{-i}^2 \end{aligned}$$

and show that this estimator converges to $\mathbb{V}_0$.

Proof. Let $\delta_i = \hat{\varphi}_{-i}(O_i) - \varphi(O_i)$. We can write

$$\begin{aligned}
\delta_i &= \left(1 - \frac{A_i}{\pi_0}\right) (\mu^*(1, X_i) - \hat{\mu}_{-i}(1, X_i)) - \left(1 - \frac{(1 - A_i)}{1 - \pi_0}\right) (\mu^*(0, X_i) - \hat{\mu}_{-i}(0, X_i)) \\
&\quad - \hat{\theta}_{\text{standardized}} + \theta_{\text{ATE}} \\
&= \left(1 - \frac{A_i}{\pi_0}\right) (\mu^*(1, X_i) - \hat{\mu}_{-i}(1, X_i)) - \left(1 - \frac{(1 - A_i)}{1 - \pi_0}\right) (\mu^*(0, X_i) - \hat{\mu}_{-i}(0, X_i)) + O_p(n^{-1/2})
\end{aligned}$$

where $\theta_{\text{ATE}} - \hat{\theta}_{\text{standardized}} = O_p(n^{-1/2})$ by (Rosenblum and van Der Laan, 2010).

We can rewrite $\hat{\varphi}_{-i}(O_i)^2 = (\varphi(O_i) - \delta_i)^2$ and thus

$$\begin{aligned}
\sqrt{n} (\hat{\mathbb{V}}_{\text{IF-LOO}} - \mathbb{V}_0) &= \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n (\varphi(O_i) - \delta_i)^2 - \mathbb{V}_0 \right) \\
&= \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n \left(\varphi(O_i)^2 - 2\varphi(O_i)\delta_i + \delta_i^2 \right) - \mathbb{V}_0 \right) \\
&= \underbrace{\sqrt{n} \left(\sum_{i=1}^n \frac{\varphi(O_i)^2}{n} - \mathbb{V}_0 \right)}_{\text{A}} - \underbrace{\frac{2}{\sqrt{n}} \sum_{i=1}^n \delta_i \varphi(O_i)}_{\text{B}} + \underbrace{\frac{1}{\sqrt{n}} \sum_{i=1}^n \delta_i^2}_{\text{C}}.
\end{aligned}$$

We'll consider the three terms separately in more detail below in each subsection. In brief, term A converges to a Normal distribution by CLT, terms B and C converge in probability to 0 as the LOO Influence function converges to the true IF in large samples.

A.1.1 Term A

By the central limit theorem, we have

$$\sqrt{n} \left(\sum_{i=1}^n \frac{\varphi(O_i)^2}{n} - \mathbb{V}_0 \right) \xrightarrow{d} \mathcal{N}(0, W),$$

where $W = \text{Var}(\varphi(O)^2) = \mathbb{E}_0(\varphi(O)^4) - \mathbb{V}_0^2$, where the fourth moment exists when condition 4 is satisfied.

A.1.2 Term C

Let $r_{-i}(a, x) = \mu^*(a, x) - \hat{\mu}_{-i}(a, x)$ be the residual of the model $\hat{\mu}_{-i}(a, x)$. Because $\left|1 - \frac{A_i}{\pi_0}\right| \leq 1 + \frac{1}{\eta}$ and $\left|1 - \frac{1-A_i}{1-\pi_0}\right| \leq 1 + \frac{1}{\eta}$ (as guaranteed by condition 2), we can apply the triangle inequality and the fact that, for real numbers a, b, c , $(a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$:

$$|\delta_i| \leq \left(1 + \frac{1}{\eta}\right) (|r_{-i}(1, X_i)| + |r_{-i}(0, X_i)|) + O_p(n^{-1/2}) \tag{9}$$

$$\delta_i^2 \leq C_1 \left((r_{-i}(1, X_i)^2 + r_{-i}(0, X_i)^2) \right) + O_p(n^{-1}), \tag{10}$$

where $C_1 = 3 \left(1 + \frac{1}{\eta}\right)^2$.

Now, consider the conditional expectation given $\{O_j : \forall j \text{ s.t. } j \neq i\}$ of both sides:

$$\begin{aligned}\mathbb{E}_0[\delta_i^2|\{O_j : j \neq i\}] &\leq C_1\left(\mathbb{E}_0[r_{-i}(1, X_i)^2|\{O_j : j \neq i\}] + \mathbb{E}_0[r_{-i}(0, X_i)^2|\{O_j : j \neq i\}]\right) + O_p(n^{-1}) \\ &\leq C_1\left(\mathbb{E}_0[r_{-i}(1, X_i)^2] + \mathbb{E}_0[r_{-i}(0, X_i)^2]\right) + O_p(n^{-1}),\end{aligned}$$

where we can drop the conditioning in the second line because, by the leave-one-out construction, $\hat{\mu}_{-i}$ is fixed given the set $\{O_j : j \neq i\}$, and O_i is independent of $\{O_j : j \neq i\}$.

Note that, because of Condition 2, we can write

$$\begin{aligned}\mathbb{E}_0[r_{-i}(1, X_i)^2]\pi_0 + \mathbb{E}_0[r_{-i}(0, X_i)^2](1 - \pi_0) &= \|\hat{\mu}_{-i} - \mu^*\|_{L_2}^2 \\ \mathbb{E}_0[r_{-i}(1, X_i)^2] + \mathbb{E}_0[r_{-i}(0, X_i)^2] &\leq \frac{1}{\eta} \|\hat{\mu}_{-i} - \mu^*\|_{L_2}^2,\end{aligned}$$

and so,

$$\mathbb{E}_0[\delta_i^2|\{O_j : j \neq i\}] \leq C_2 \|\hat{\mu}_{-i} - \mu^*\|_{L_2}^2,$$

where $C_2 = C_1/\eta$.

By triangle inequality, we have

$$\|\hat{\mu}_{-i} - \mu^*\|_{L_2} \leq \underbrace{\|\hat{\mu}_{-i} - \hat{\mu}\|_{L_2}}_I + \underbrace{\|\hat{\mu} - \mu^*\|_{L_2}}_{II},$$

where $\hat{\mu}$ is the logistic regression fit on the whole data set $O_1, \dots, O_n$. By condition 3, term I = $o_p(n^{-1/2})$. And by condition 1, if the regressions are fit using maximum likelihood estimation, then term II = $O_p(n^{-1/2})$ (Vaart, 1998). Thus,

$$\|\hat{\mu}_{-i} - \mu^*\|_{L_2}^2 = O_p(n^{-1}).$$

Since this holds uniformly over all i , we can take the average:

$$\begin{aligned}\frac{1}{n} \sum_{i=1}^n \mathbb{E}_0[\delta_i^2|\{O_j : j \neq i\}] &\leq C_2 O_p(n^{-1}) + O_p(n^{-1}) \\ &= o_p(n^{-1/2})\end{aligned}$$

Now, we consider the term $\left| \frac{1}{n} \sum_{i=1}^n \delta_i^2 - \frac{1}{n} \sum_{i=1}^n \mathbb{E}_0[\delta_i^2|\{O_j : j \neq i\}] \right|$, the magnitude of the difference between the sum and the conditional mean. First, note that δ_i and δ_j are independent for $i \neq j$. Thus, we can write

$$\begin{aligned}\text{Var} \left(\frac{1}{n} \sum_{i=1}^n \delta_i^2 \middle| \{O_j : j \neq i\} \right) &= \frac{1}{n^2} \sum_{i=1}^n \text{Var} \left(\delta_i^2 \middle| \{O_j : j \neq i\} \right) \\ &\leq \frac{1}{n^2} \sum_{i=1}^n \mathbb{E}_0 \left[\delta_i^4 \middle| \{O_j : j \neq i\} \right]\end{aligned}$$

By Equation 10,

$$\begin{aligned}\delta_i^4 &\leq C_1^2 \left((r_{-i}(1, X_i)^2 + r_{-i}(0, X_i)^2)^4 \right) \\ &\leq 8C_1^2 \left((r_{-i}(1, X_i)^4 + r_{-i}(0, X_i)^4) \right)\end{aligned}$$

The same argument as above gives

$$\mathbb{E}_0[\delta_i^4 | \{O_j : j \neq i\}] \leq 8C_2 \|\hat{\mu}_{-i} - \mu^*\|_{L_2}^4 = O_p(n^{-2}). \quad (11)$$

Thus,

$$\begin{aligned}\text{Var} \left(\frac{1}{n} \sum_{i=1}^n \mathbb{E}_0[\delta_i^2 | \{O_j : j \neq i\}] \right) &\leq \frac{1}{n^2} \sum_{i=1}^n \mathbb{E}_0[\delta_i^4 | \{O_j : j \neq i\}] \\ &= \frac{1}{n^2} \times n \times O_p(n^{-2}) \\ &= O_p(n^{-3}).\end{aligned}$$

By Chebyshev's inequality, for any $\epsilon > 0$, we have

$$\begin{aligned}\mathbb{P}_0 \left(\left| \frac{1}{n} \sum_{i=1}^n \delta_i^2 - \frac{1}{n} \sum_{i=1}^n \mathbb{E}_0[\delta_i^2 | \{O_j : j \neq i\}] \right| > \epsilon n^{-1/2} \middle| \{O_j : j \neq i\} \right) &\leq \frac{\text{Var} \left(\frac{1}{n^2} \sum_{i=1}^n \mathbb{E}_0[\delta_i^2 | \{O_j : j \neq i\}] \right)}{\epsilon^2 n^{-1}} \\ &= \frac{O_p(n^{-2})}{\epsilon^2} \\ &\rightarrow 0.\end{aligned}$$

Therefore, by applying law of total probability, we have shown

$$\left| \frac{1}{n} \sum_{i=1}^n \delta_i^2 - \frac{1}{n} \sum_{i=1}^n \mathbb{E}_0[\delta_i^2 | \{O_j : j \neq i\}] \right| = o_p(n^{-1/2}).$$

We can then write

$$\begin{aligned}\frac{1}{n} \sum_{i=1}^n \delta_i^2 &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}_0[\delta_i^2 | \{O_j : j \neq i\}] + \frac{1}{n} \sum_{i=1}^n \delta_i^2 - \frac{1}{n} \sum_{i=1}^n \mathbb{E}_0[\delta_i^2 | \{O_j : j \neq i\}] \\ &= o_p(n^{-1/2})\end{aligned}$$

and thus

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \delta_i^2 = o_p(1).$$

A.1.3 Term B

Because of leave one out, O_i and $\hat{\mu}_{-i}$ are independent conditional on $\{O_j : j \neq i\}$. Thus, we have

$$\mathbb{E}_0[\varphi(O_i) \delta_i | \{O_j : j \neq i\}] = 0.$$

By conditional independence of $\varphi(O_i)\delta_i$ over $i = 1, \dots, n$, we can write

$$\begin{aligned} \text{Var} \left(\frac{2}{\sqrt{n}} \sum_{i=1}^n \varphi(O_i)\delta_i \middle| \{O_j : j \neq i\} \right) &= \frac{4}{n} \sum_{i=1}^n \mathbb{E}_0 \left[\varphi(O_i)^2 \delta_i^2 \middle| \{O_j : j \neq i\} \right] \\ &\leq \frac{4}{n} \mathbb{E}_0 \left[\varphi(O_i)^4 \middle| \{O_j : j \neq i\} \right]^{1/2} \mathbb{E}_0 \left[\delta_i^4 \middle| \{O_j : j \neq i\} \right]^{1/2} \\ &= \frac{4}{n} \mathbb{E}_0 \left[\varphi(O_i)^4 \right]^{1/2} \mathbb{E}_0 \left[\delta_i^4 \middle| \{O_j : j \neq i\} \right]^{1/2}, \end{aligned}$$

where the inequality is by Cauchy-Schwarz, and the equality in line 3 is due to the independence of O_i across $i = 1, \dots, n$. By Condition 4, $\mathbb{E}_0 [\varphi(O_i)^4]^{1/2}$ is finite. For the second term, we apply the same pointwise bound as in Equation 11. Thus,

$$\text{Var} \left(\frac{2}{n} \sum_{i=1}^n \varphi(O_i)\delta_i \middle| \{O_j : j \neq i\} \right) = \frac{4}{n} \times n \times O_p(n^{-1}) = O_p(n^{-1}).$$

Then, we can apply Chebyshev's inequality. For any $\epsilon > 0$,

$$\begin{aligned} \mathbb{P}_0 \left(\left| \frac{2}{\sqrt{n}} \sum_{i=1}^n \varphi(O_i)\delta_i \right| > \epsilon \middle| \{O_j : j \neq i\} \right) &\leq \frac{O_p(n^{-1})}{\epsilon^2} \\ &\rightarrow 0. \end{aligned}$$

Therefore,

$$\frac{2}{\sqrt{n}} \sum_{i=1}^n \varphi(O_i)\delta_i = o_p(1).$$

A.1.4 Combining all the terms

Combining terms and applying Slutsky's theorem,

$$\begin{aligned} \sqrt{n} \left(\hat{\mathbb{V}}_{\text{IF-LOO}} - \mathbb{V}_0 \right) &= \underbrace{\sqrt{n} \left(\sum_{i=1}^n \frac{\varphi(O_i)^2}{n} - \mathbb{V}_0 \right)}_{\xrightarrow{d} \mathcal{N}(0, W)} - \underbrace{\frac{2}{\sqrt{n}} \sum_{i=1}^n \delta_i \varphi(O_i)}_{o_p(1)} + \underbrace{\frac{1}{\sqrt{n}} \sum_{i=1}^n \delta_i^2}_{o_p(1)} \\ &\xrightarrow{d} \mathcal{N}(0, W). \end{aligned}$$

Thus, $\mathbb{V}_{\text{IF-LOO}}$ is asymptotically normal and $\sqrt{n}$ consistent with asymptotic variance $W = \text{Var}(\varphi(O)^2)$.

B Data Generating Processes for Simulation Studies

This section describes the data-generating process used in Section 3. All of the simulations presented in this paper have data generating processes of the same structure:

$$X_1, X_2, X_3, X_4 \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1)$$

$$\begin{aligned}
X_5, X_6 &\stackrel{\text{iid}}{\sim} \text{Bernoulli}(p = 1/2) \\
A &\sim \text{Bernoulli}(p = 1/2) \\
Y|A, \mathbf{X} &\sim \text{Bernoulli}(p = \text{expit}(\beta_0 + \beta_A A + \beta_{\mathbf{X}} \mathbf{X})) \\
\mathbf{X} &= [X_1, X_2, X_3, X_4, X_5, X_6]^T \\
\beta_{\mathbf{X}} &= [2.5, 1.8, -2.8, -2.1, 2.0, -2.0]
\end{aligned}$$

Table 3 describes the other parameters that change throughout the two scenarios presented:

Table 3: Data Generating Process Parameters

Placebo Rate	N	β_0	ATE	β_A
2.5%	250	-1.4828	0%	0
			2.5%	0.8566
			5%	1.4065
			10%	2.1752
			15%	2.7465
25%	50	-4.9171	0%	0
			2.5%	0.2028
			5%	0.3967
			10%	0.7643
			15%	1.1131